

10 LANGCHAIN CONCEPTS EVERY AI ENGINEER SHOULD KNOW



LangChain

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WHAT IS LANGCHAIN

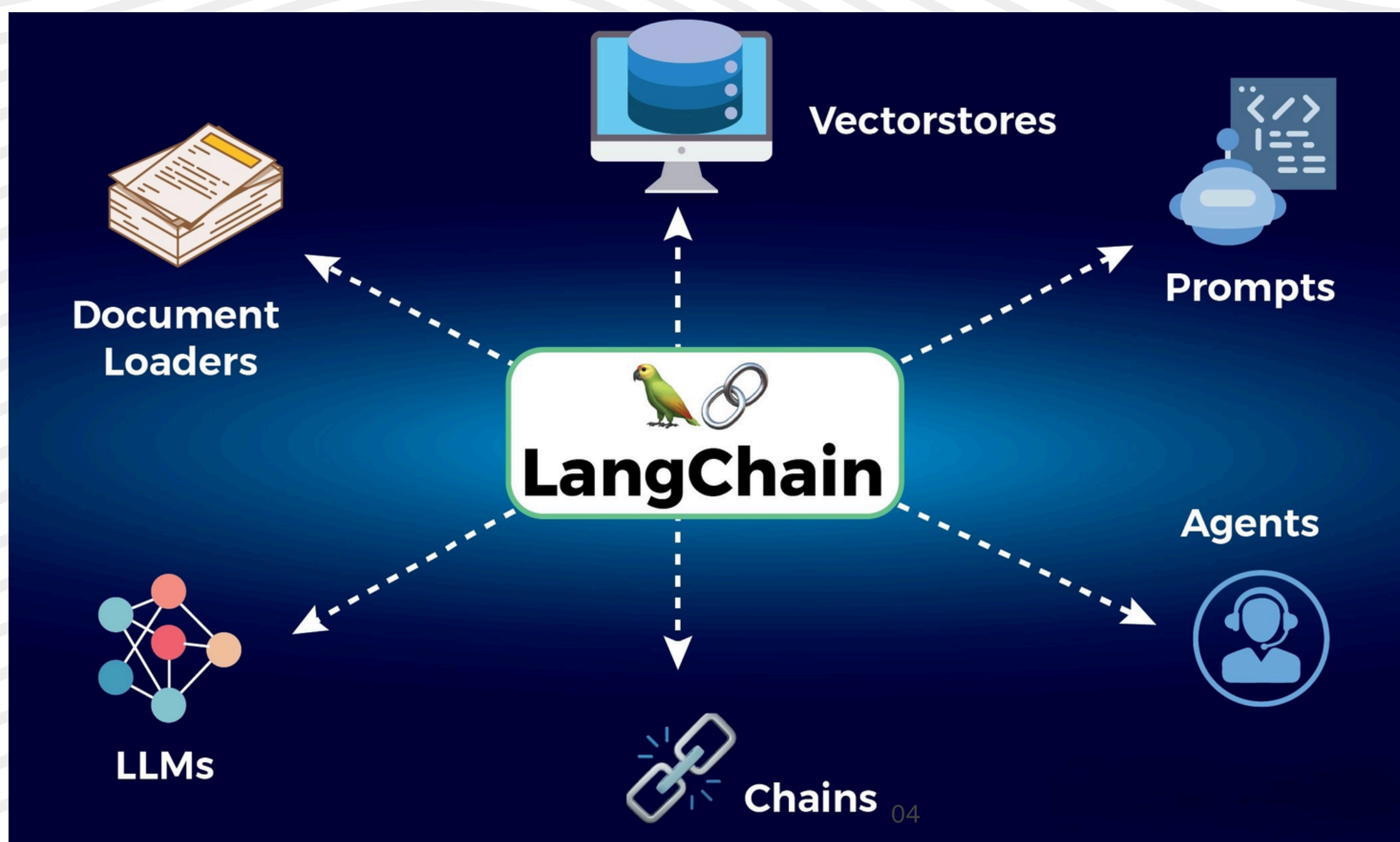
The Framework for LLM Applications

LangChain helps developers build applications powered by Large Language Models (LLMs).

It provides reusable components to connect:

- LLMs
- Databases
- APIs
- Documents
- External Tools

Instead of calling an LLM directly, LangChain helps create complete AI workflows.



MODELS

The Brain of Every LangChain Application

Models are the core AI engines that generate responses.

LangChain supports:

- OpenAI GPT Models
- Anthropic Claude
- Google Gemini
- Open Source LLMs



```
llm.invoke("Explain Data  
Warehousing")
```

The model processes input and generates output.

PROMPT TEMPLATES

Reusable Prompt Engineering for Scalable AI Applications

Instead of hardcoding prompts, LangChain allows you to create dynamic templates that accept user inputs.



```
from langchain_core.prompts import PromptTemplate

prompt = PromptTemplate.from_template(
    "Explain {topic} in simple terms."
)

prompt.invoke({"topic": "Data Warehousing"})
```

Reusable Prompt Engineering for Scalable AI Applications

Instead of hardcoding prompts, LangChain allows you to create dynamic templates that accept user inputs.

CHAINS

Connecting Multiple Steps Together

A Chain links multiple operations into a workflow. Instead of one API call, chains create structured AI pipelines

Benefits:

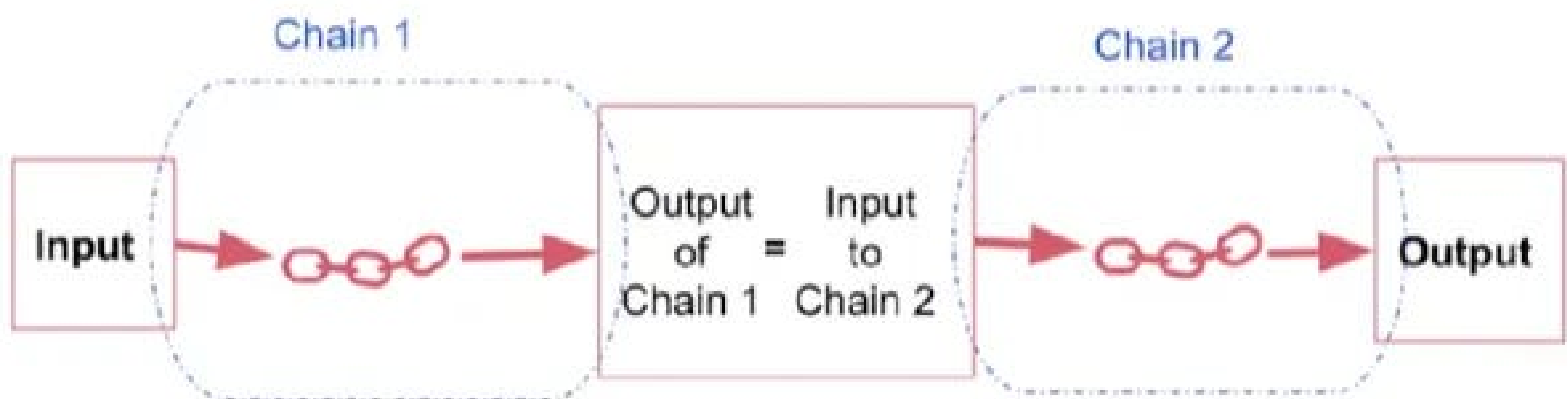
- Modular Design
- Reusable Logic
- Cleaner Applications

User Question

Prompt Template

LLM

Output



MEMORY

Giving AI Context

By default, LLMs forget previous interactions. Memory allows AI applications to remember:

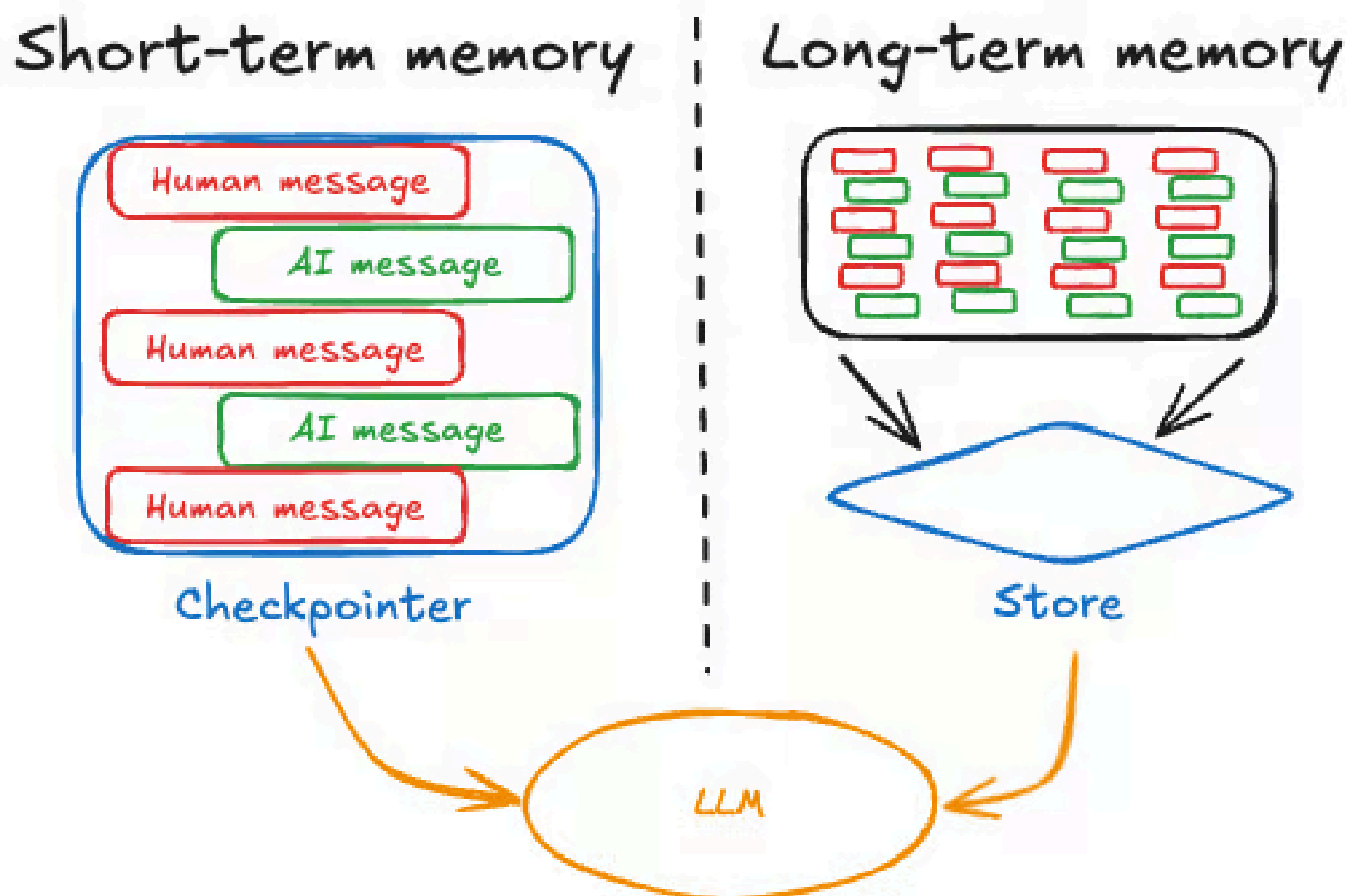
- User Preferences
- Previous Questions
- Conversation History
- Session Context

Example:

User: "My favorite fruit is Mango."

Later:

AI remembers and responds accordingly.

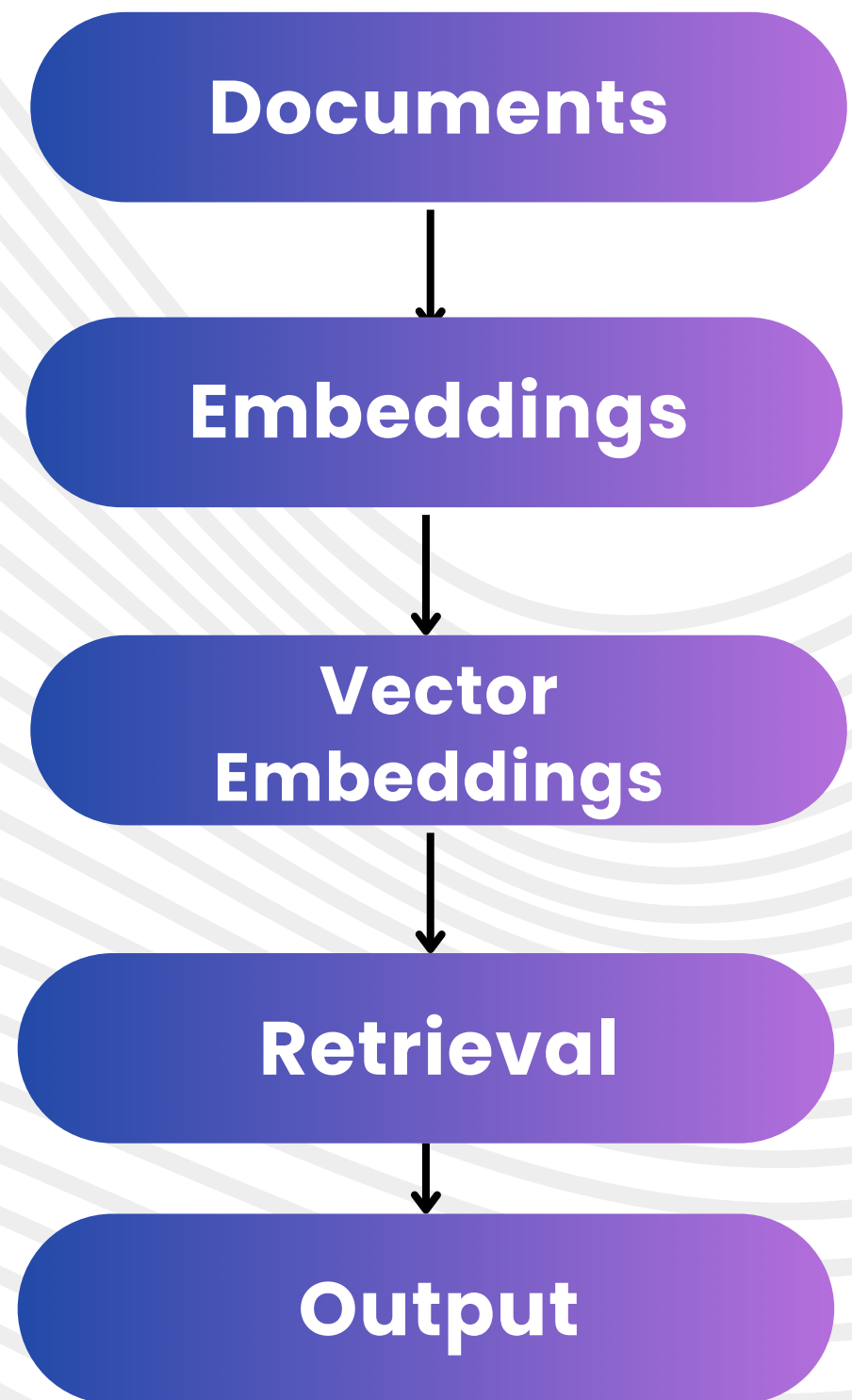


RETRIEVAL-AUGMENTED GENERATION (RAG)

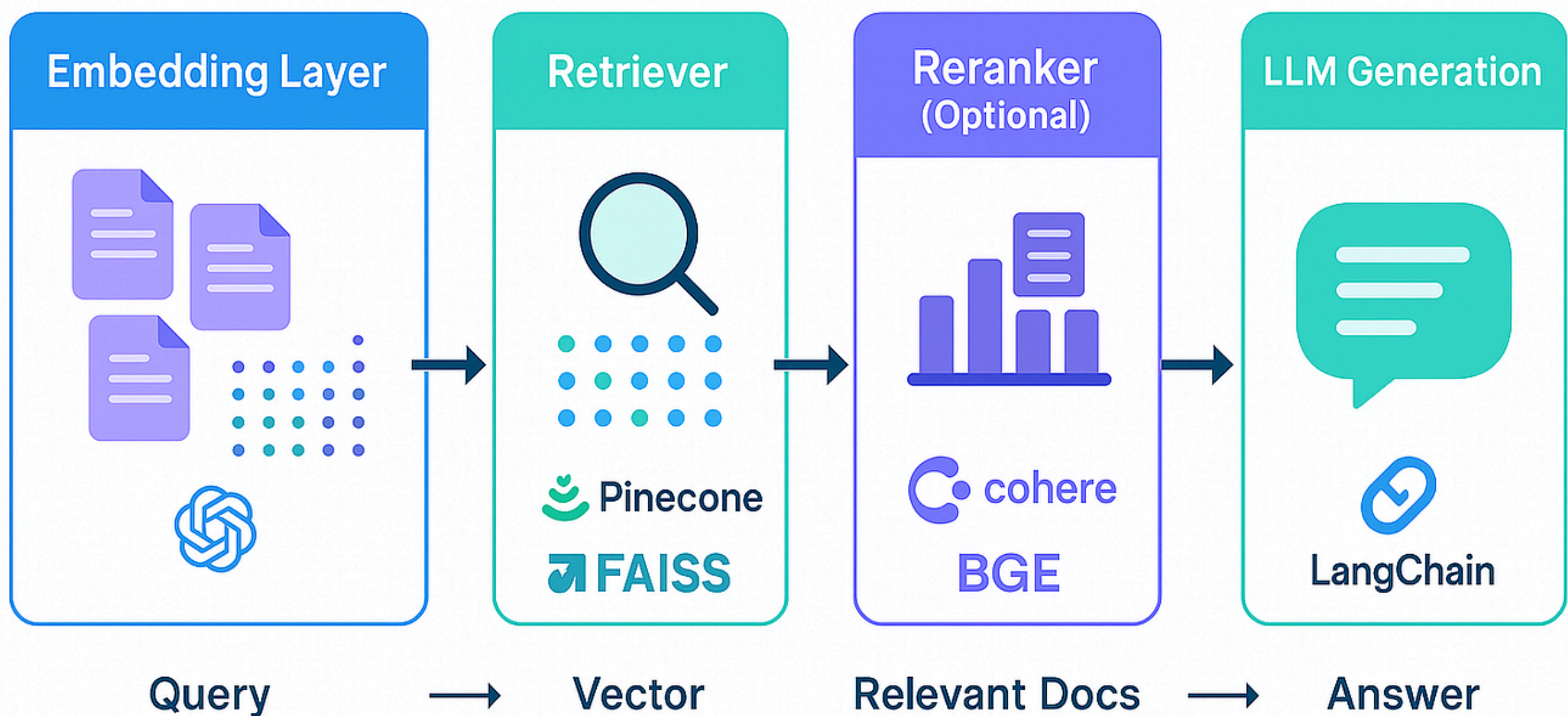
LLMs don't know your data or company documents.

RAG solves this problem.

- More Accurate Answers
- Reduced Hallucinations
- Access to Private Data
- Real-Time Knowledge



LangChain RAG Architecture



VECTOR DATABASES

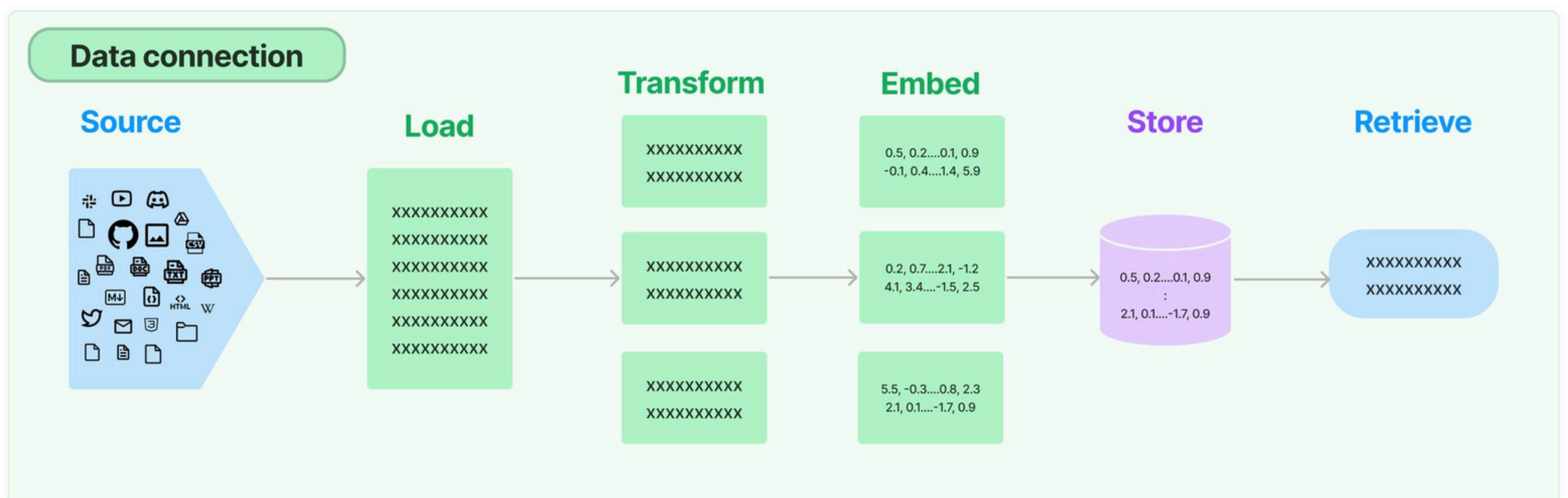
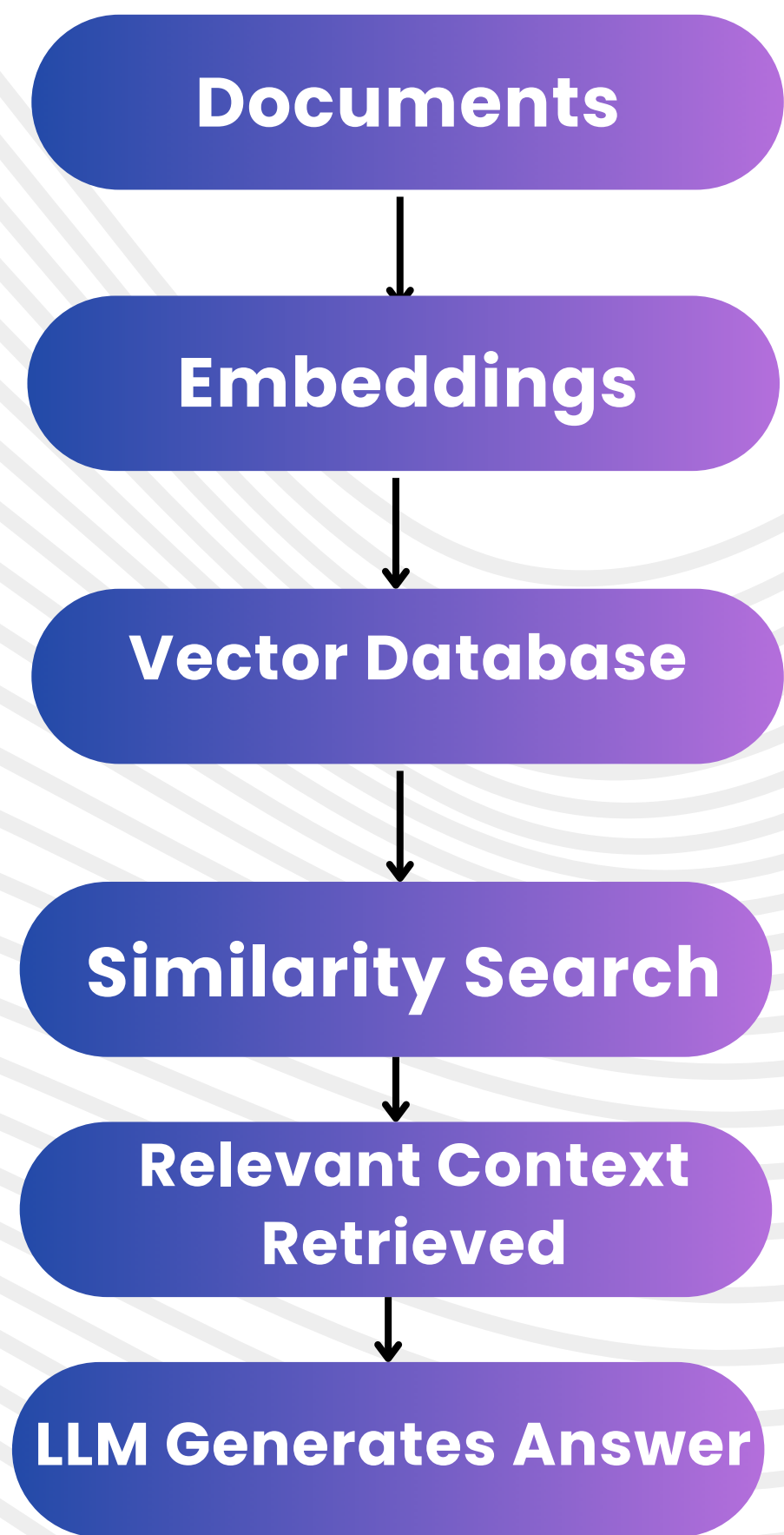
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Popular Vector Databases

- FAISS
- Pinecone
- Chroma
- Weaviate
- Mivus



TOOLS & AGENTS

Tools allow LLMs to interact with external systems.

Agents decide:
"What tool should I use?"

Example:

User: "What's the weather in Mumbai?"

Agent:

1. Chooses Weather API
2. Calls Tool
3. Returns Answer

This transforms chatbots into autonomous assistants.



MASTER THESE 10 CONCEPTS:

- 1. MODELS**
- 2. PROMPT TEMPLATES**
- 3. CHAINS**
- 4. MEMORY**
- 5. RAG**
- 6. EMBEDDINGS**
- 7. VECTOR STORES**
- 8. TOOLS**
- 9. AGENTS**
- 10. LANGGRAPH**

**LEARN THESE CONCEPTS, AND YOU'LL
UNDERSTAND HOW MOST MODERN AI
APPLICATIONS ARE BUILT.**

The background of the image consists of numerous thin, light gray lines that flow in a wavy, undulating pattern across the entire frame. These lines create a sense of movement and depth, resembling a stylized topographical map or a liquid surface. The lines are more densely packed in some areas and more spread out in others, contributing to the overall organic feel of the design.

THANK YOU